3.2 $Y=(B\cdot X)/(1-p)$,

$B_{ij}\sim Bernoulli(1-p)$

$B\in\{0,1\}^{d\times m}$

4. $L=L(Y)\in R$, $\frac{dL}{dY}\Rightarrow\frac{dL}{dX}$

solve: $\frac{dL}{dX}=\frac{B\cdot\frac{dL}{dY}}{1-p}$, ($\cdot$ is $\dot{e}$)

dropout

1. $y=f(x)$, $x,y\in R^d$

1.2 $y=\frac{b}{1-p}x$, mask $b\sim Bernoulli(1-p)$, $b\in R^d$

p is dropout probability, $p=0.1$.

here $y_i=\begin{cases}0, & \text{if } b_i=0\\ \frac{x_i}{1-p}, & \text{if } b_i=1\end{cases}$

1.3 $E(y)=\frac{E(b)}{1-p}x=\frac{1-p}{1-p}x=x$

1.4 $f(x)=x$ when inference

2. $L=L(y)\in R$, $\frac{dL}{dy}\Rightarrow\frac{dL}{dx}$

($\frac{dL}{dX}:=(\frac{\partial L}{\partial X}).reshape(X.shape)$, $\forall X$)

solve: $\frac{dL}{dx}=\frac{b}{1-p}\cdot\frac{dL}{dy}$ ($\cdot$ is $\dot{e}$)

3. $Y=f(X)$, $X,Y\in R^{d\times m}$

d indexes features, m indexes examples.